Climate Resilience as a Housing Market Signal: Evidence from NOAA Storm Events and FEMA NRI Data

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Contents

1 Abstract **2**

2 Motivation **3**

3 Data **3**
3.1 Data Sources . . . 3
3.2 Sample Construction . . . 3
3.3 Exploratory Analysis . . . 4

4 Method **4**
4.1 Prior Research . . . 4
4.2 Event Study Design . . . 5
4.3 Cumulative Impulse Response . . . 5
4.4 Regression Specification . . . 5

5 Results **6**
5.1 Event Study . . . 6
5.2 Regression Results . . . 7
5.3 Local Projections . . . 8

6 Discussion **9**
6.1 Interpretation . . . 9
6.2 Future Work . . . 10

7 Appendix **11**

8 References **13**

1 Abstract

To be completed.

2 Motivation

To be completed.

3 Data

3.1 Data Sources

We combine three datasets at the county-month level:

NOAA Storm Events Database (2020–2025). Event-level records of storm occurrences across the United States, including location, event type, duration, and damage estimates. We aggregate to county-month observations. Our sample covers 8,648 county-month storm observations across 2,180 unique counties over 2020-2025.

Zillow Home Value Index (ZHVI). Monthly county-level median home values for single-family residences and condominiums, seasonally adjusted, representing the 35th to 65th percentile of the housing market.

FEMA National Risk Index (NRI). County-level scores for hazard exposure, social vulnerability, and community resilience. We match each storm observation to the most recent available NRI vintage. Our primary variable of interest is `resl_score`, a national percentile ranking of each county's capacity to prepare for, absorb, and recover from natural hazards.

3.2 Sample Construction

We retain storm events with complete 9-month post-storm and 3-month pre-storm ZHVI windows. Events are aggregated at the count-month level. Events after May 2025 are dropped due to insufficient post-storm coverage. Alaska, Hawaii, and Puerto Rico are excluded. Final analytic sample: 8,648 county-month observations.

3.3 Exploratory Analysis

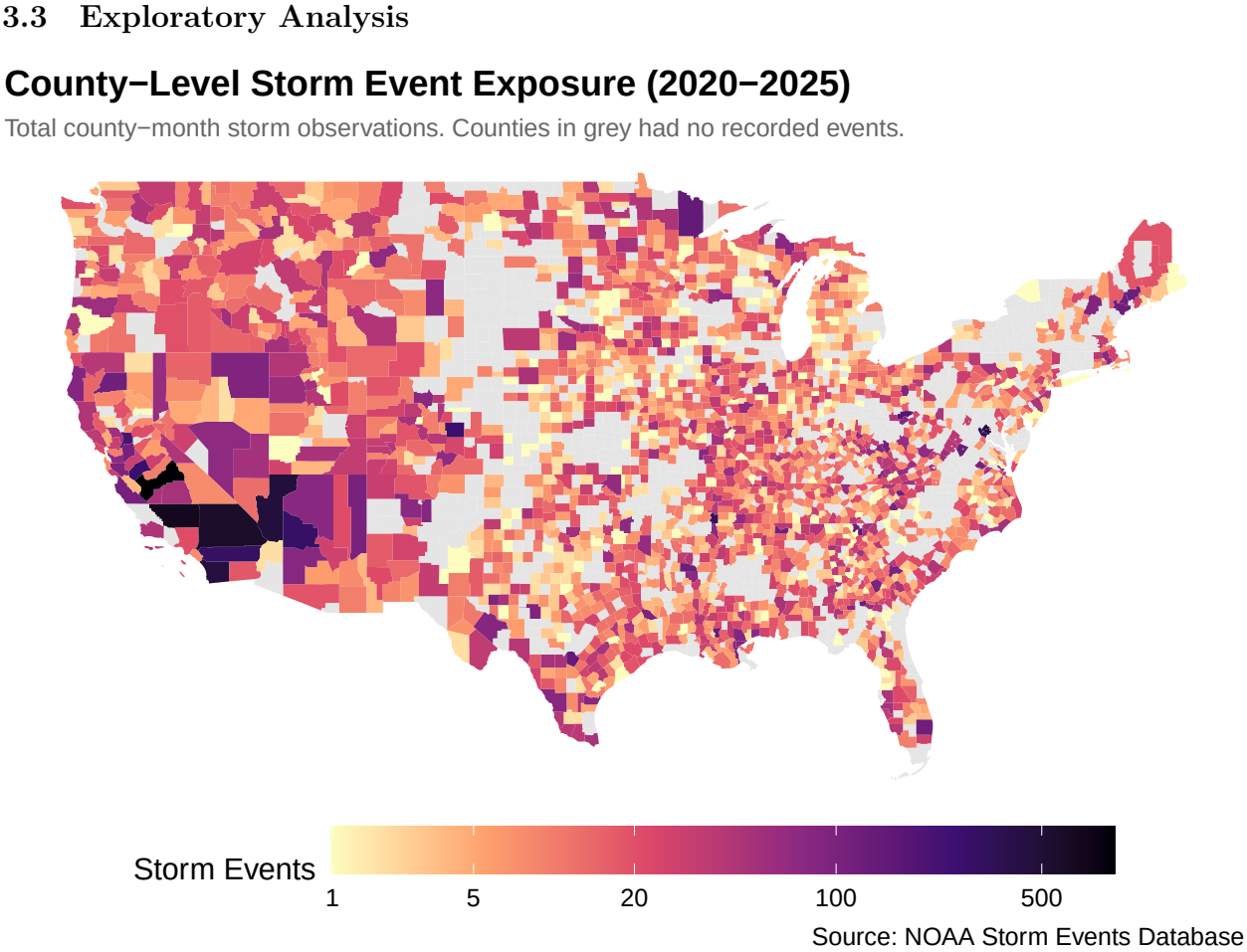

Figure 1: County-level storm exposure across the contiguous United States, 2020 to 2025. Counties shaded grey recorded no storm events in the sample period.

Figure 1 shows the geographic distribution of storm exposure. Storm activity is concentrated along the Gulf Coast, the Southeast, and Tornado Alley - consistent with known patterns of hurricane, flood, and severe convective storm activity. The Midwest and Plains states show substantial exposure to storm events in the database.

4 Method

4.1 Prior Research

Our methodology is grounded in the Local Projections Difference in Differences (LP DiD) work of Dube, Girardi, Jorda' and Taylor (2025). LP-DiD uses a local projects framework with difference-in-differences design to measure the effect of a shock to the system over a post-treatment window of time to capture the full impact of the shock. The estimator considers both the magnitude of the response to an economic shock and the duration of the aftereffects. Using multiple time based regressions, the LP framework estimates a time series curve for all periods after the shock event that are summed to a single impulse response representing the the total post-shock effect.

We make a simplifying assumption in this study to combine the post-storm effects into a single outcome before analysis. The tradeoff is that our estimate loses precision in estimating the time and magnitude of the shock as separate quantities. For the consideration of climate resilience we care about both magnitude

of price disruption and duration of recovery as both meaningful outcomes of storm events. The outcome of measure (Cumulative Impulse Response) accumulates the effect magnitude over time into a single measurable outcome.

4.2 Event Study Design

For each county c affected by a storm, we examine ZHVI before and after the event. All county ZHVI ($Z_{c,t}$) are rebased to 100 at $t = 0$, so deviations represent percentage changes from the storm month.

For each storm event at county c and time t, we construct a neighbor baseline from adjacent counties that recorded no storm events during the full ± 3 to $+9$ month window. The baseline $B_{c,t+n}$ is the equal-weighted mean ZHVI of these clean neighbor counties, indexed to 100 at $t = 0$. The monthly deviation is:

$$\delta_{c,t+n} = Z_{c,t+n} - B_{c,t+n}$$

4.3 Cumulative Impulse Response

Cumulative Impulse Response measures the total area under the curve of the deviations $\delta_{c,t}$ to capture both magnitude and time effects in one measurement.

$$\text{CIR}_{c,t} = \sum_{n=1}^{9} \delta_{c,t+n}$$

The pre-storm trend annualizes the average monthly deviation over the 3 months prior to the storm:

$$\text{PreTrendAnnual}_{c,t} = 12 \times \frac{1}{3} \sum_{n=1}^{3} \delta_{c,t-n}$$

4.4 Regression Specification

We estimate a series of two-way fixed effects models of the form:

$$\text{CIR}_{c,t} = \beta_1 \cdot \text{PreTrendAnnual}_{c,t} + \mathbf{X}'_{c,t}\gamma + \alpha_c + \lambda_t + \varepsilon_{c,t}$$

where α_c are county fixed effects, λ_t are month-year fixed effects, $\mathbf{X}_{c,t}$ is a vector of storm severity and resilience controls, and standard errors are two-way clustered at the county and month-year level.

County fixed effects absorb time-invariant county characteristics - geography, long-run economic conditions, and local housing market features. Month-year fixed effects absorb common macroeconomic shocks including national interest rate movements and aggregate housing demand.

Our preferred specification (M5) augments the baseline with an interaction between log property damage and resilience quartile, testing whether the damage-to-price relationship differs across the resilience distribution.

We complement the regression analysis with local projections (Jordà, 2005) that estimate the dynamic path of the damage and resilience interaction effects at each post-storm horizon $t + 1$ through $t + 9$.

5 Results

5.1 Event Study

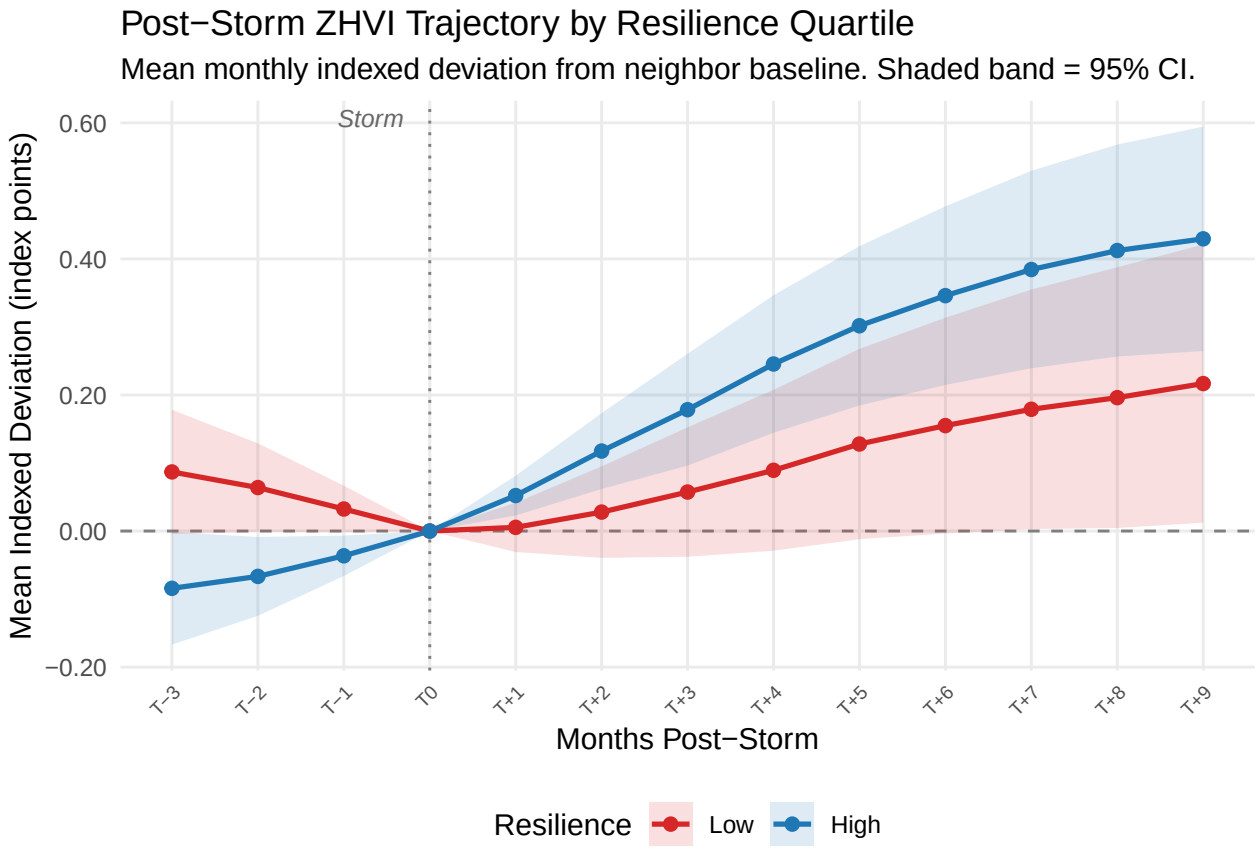

Figure 2: Average post-storm ZHVI trajectory for high resilience (top quartile) and low resilience (bottom quartile) counties. Shaded bands represent 95% confidence intervals.

Key observations:

- Both high and low resilience counties show post-storm price appreciation relative to neighbor baseline
- Event type trajectories are broadly similar across categories, suggesting storm type alone does not drive recovery heterogeneity
- Pre-storm deviations are non-zero, motivating `pre_trend_annual` as a control variable

5.2 Regression Results

Dependent Variable:	Cumulative Impulse Response (CIR)				
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Pre-Storm Trend (Annual)	-0.3423***	-0.3426***	-0.3384***	-0.3381***	-0.3374***
	(0.0401)	(0.0400)	(0.0398)	(0.0397)	(0.0393)
Event Count		0.0268	0.0264	0.0222	0.0171
		(0.0194)	(0.0188)	(0.0194)	(0.0193)
Log Property Damage		-0.0494	-0.0520	0.3878*	-0.2310**
		(0.0616)	(0.0613)	(0.1953)	(0.0924)
Resilience Score			0.0884**	0.1170***	
			(0.0390)	(0.0422)	
Log Property Damage × Resilience Score				-0.0083**	
				(0.0034)	
Resiliency Low (Q1)					-3.051
					(2.595)
Resiliency High (Q4)					2.762
					(2.324)
Log Property Damage × Resiliency Low (Q1)					0.5863***
					(0.1732)
Log Property Damage × Resiliency High (Q4)					0.1714
					(0.1517)
Fixed-effects					
month_year	Yes	Yes	Yes	Yes	Yes
stcofips	Yes	Yes	Yes	Yes	Yes
Fit statistics					
Observations	8,172	8,172	8,172	8,172	8,172
R^2	0.46773	0.46782	0.46906	0.46970	0.47006
Within R^2	0.06279	0.06295	0.06514	0.06626	0.06690

Clustered (stcofips & month_year) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Key findings:

- Pre-storm trend is the strongest predictor of CIR across all specifications - mean reversion is robust ($\hat{\beta} = -0.342$, $p < 0.001$)
- Storm severity controls (event count, log damage) are not independently significant after absorbing county and time fixed effects
- Resilience score is positively associated with CIR after controlling for storm severity (M3: $\hat{\beta} = 0.088$, $p < 0.05$) - more resilient counties outperform their neighbors post-storm
- Once damage is interacted with resilience (M4), log damage flips positive ($\hat{\beta} = 0.388$) and the interaction is negative ($\hat{\beta} = -0.008$, $p < 0.05$)
- M5 reveals the mechanism is concentrated in Low resilience counties - damage drives anomalous appreciation relative to Middle ($\hat{\beta} = 0.586$, $p < 0.001$), while High quartile counties are statistically indistinguishable from Middle ($\hat{\beta} = 0.171$, $p > 0.1$)

5.3 Local Projections

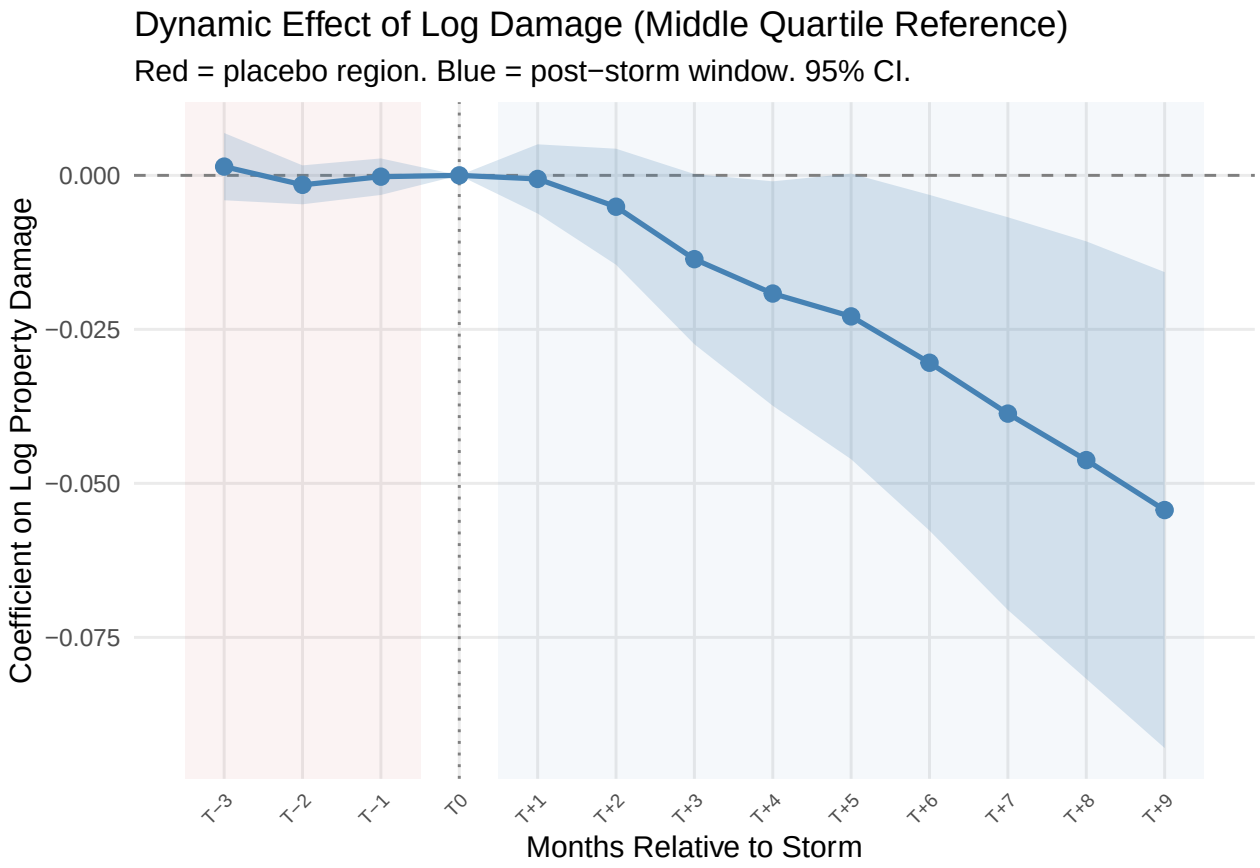

Figure 3: Dynamic effect of log property damage...

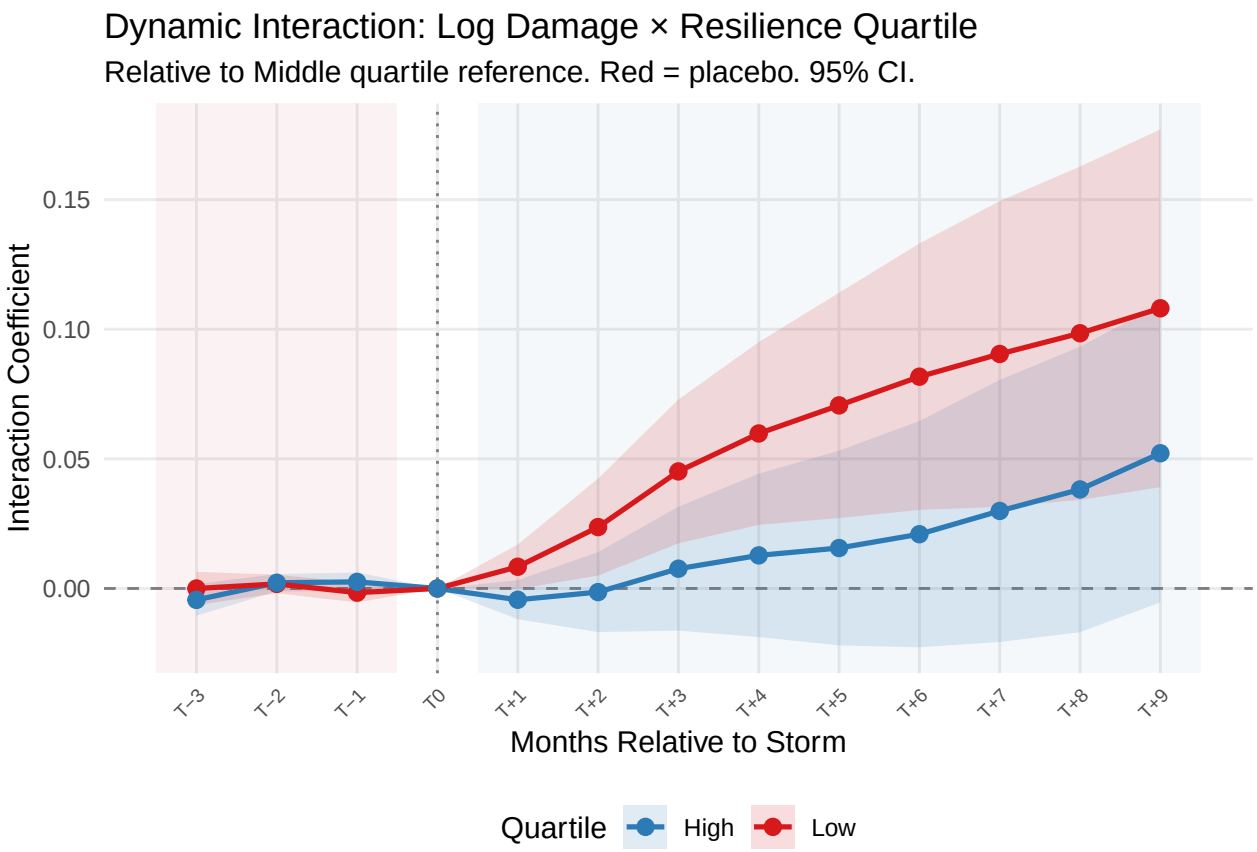

Figure 4: Dynamic interaction between log property damage...

Key findings:

- Placebo region (T-3 to T-1) is flat for all coefficients - supports identification assumption that damage effects are not pre-existing
- Damage effect for Middle quartile counties turns negative post-storm and stabilizes by T+4 to T+5 - price depreciation is persistent within the observation window
- Low resilience counties show consistently positive damage interaction throughout the post-storm window - anomalous appreciation relative to Middle emerges immediately at T+1 and persists
- High resilience counties show a smaller, noisier positive interaction - statistically indistinguishable from Middle at most horizons
- Effects appear largely baked in by T+5 to T+6, suggesting the housing market price response to storm damage is complete within approximately six months

6 Discussion

6.1 Interpretation

Key findings and open questions for the team:

- **Mean reversion dominates** - pre-storm trend is the strongest predictor of post-storm CIR across all specifications.
- **Resilience matters but the mechanism is unclear** - high resilience counties outperform on average (M3), but the interaction results (M4, M5) suggest the channel runs through damage response rather than a direct resilience effect. Why?
- **Low resilience counties show anomalous damage-driven appreciation** - why?

  - Supply shock?
  - Reconstruction?
  - Economic repercussions?
- **Price appreciation everywhere in raw trajectories** - the unconditional event study plots show appreciation across all groups.

6.2 Future Work

To be completed.

7 Appendix

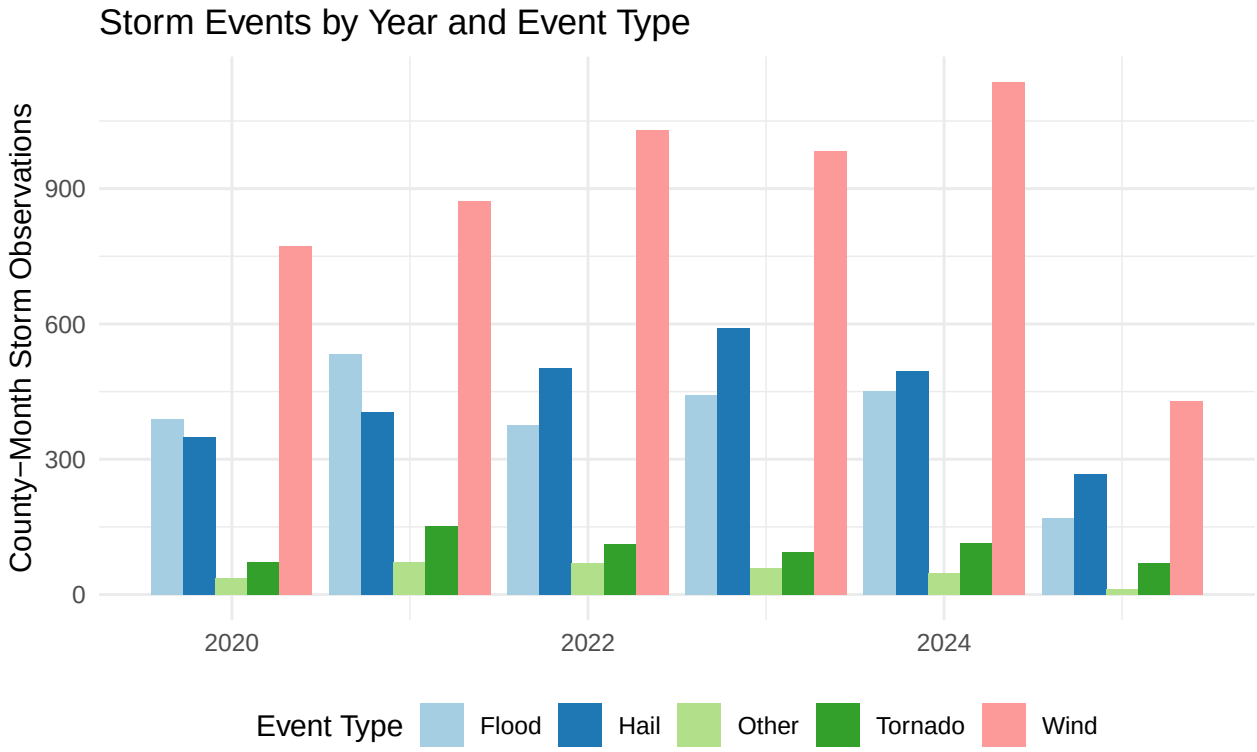

Figure 5: County-month storm observations by year and event type category.

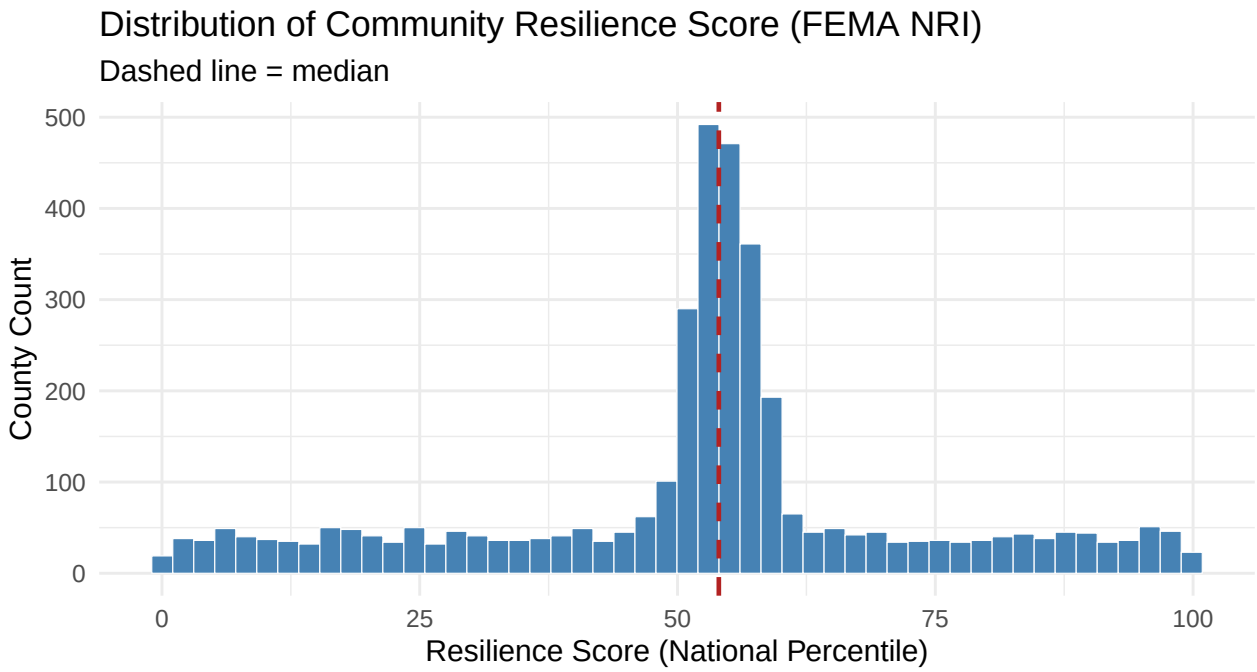

Figure 6: Distribution of FEMA NRI community resilience scores across storm-affected counties. Dashed line indicates the median resilience score of 53.9.

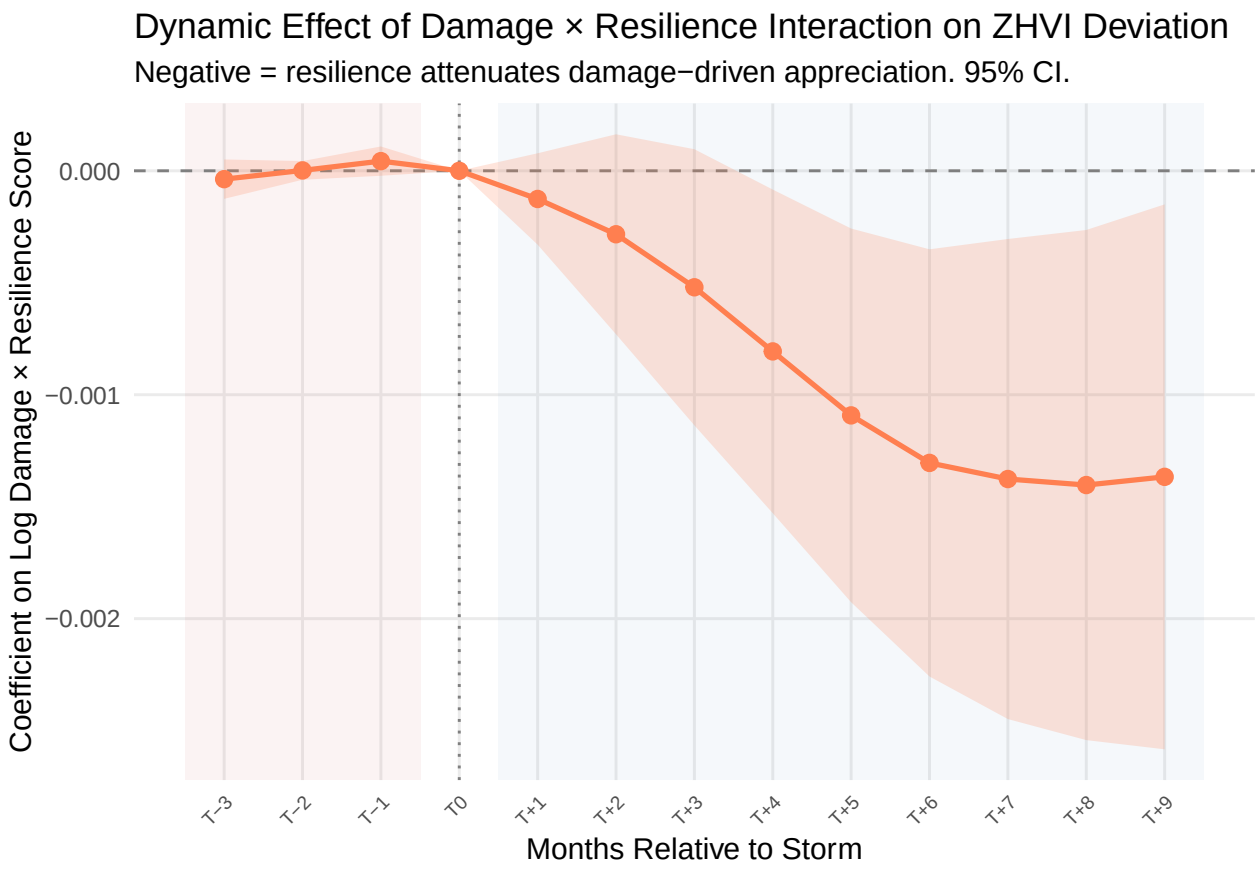

Figure 7: Dynamic effect of property damage x resilience on ZHVI deviation at each post-storm horizon.

8 References

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